Supplementary Material to "Sparse Quantum-Fuzzy Logical Petri Nets: A Compressed Sparse Row Benchmark"

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This file collects material that supports, but is not needed to follow, the main article: the comparison matrix of the planned experiments, the numerical consistency check, the artifact definitions and dependency structure, the registries of benchmark quantities and artifacts, and a template for reporting confidence intervals. None of its tables reports a new result.

1 Comparison Matrix for Future Benchmark Experiments

The comparisons must use the same mathematical problem whenever the independent variable is not itself the problem definition. A sequential-versus-parallel comparison is invalid if the two implementations use different sparse matrices or different input vectors. Similarly, a threshold experiment is meaningful only when the matrix, vector, hardware and thread count remain fixed.

2 Numerical Consistency Audit

The speedup is computed consistently from the displayed benchmark times:

$$S = \frac{T_{\text{seq}}}{T_{\text{par}}} = \frac{8.250}{3.139} \approx 2.6282. \tag{1}$$

Accordingly, this manuscript reports $2.6282\times$ throughout when referring to the ratio of the displayed times. Because the underlying raw timing values are not supplied in this manuscript, no more precise claim is made about a ratio computed from unrounded measurements.

3 Artifact Definition and Benchmark Dependency Structure

3.1 Definition: Reproducible Artifact

A reproducible artifact is a source file, input dataset, configuration, script, raw measurement record, or output that can be reconstructed from the declared procedure and dependencies.

Table 1: Controlled comparisons proposed for the experimental campaign.

Comparison	Primary measurements
Dense vs. sparse representation	memory requirement, runtime, numerical agreement
Sequential vs. parallel	thread count, runtime, speedup, error
Threshold ablation	task threshold, runtime, task count, error
Pattern sensitivity	row-length distribution, runtime, load-balance metric, error
Dimension scaling	N and NNZ, runtime, memory, speedup, failure point

Table 2: Computational dependency structure for the benchmark.

Result	**Prerequisite / dependency**
CSR correctness	Valid row pointers, column indices, and value/index correspondence
Parallel row independence	Disjoint row ownership and no shared output writes
Sparse storage complexity	Declared CSR representation and dimensions N, NNZ
Runtime measurement	Timing procedure, configuration, and recorded observations
Speedup	Sequential and parallel runtimes measured for a comparable workload
Statistical confidence interval	Multiple independent raw timing observations and a declared estimator

Listing a future artifact does not mean that it is already deposited or publicly accessible. For benchmark reproduction, the artifact should include raw timings and the hardware/software environment, not only the displayed mean or selected runtime.

3.2 Definition: Benchmark Reproducibility

A benchmark is reproducible to the extent that its workload dimensions, nonzero count, sparse representation, thread configuration, hardware/software environment, warm-up policy, timing procedure, correctness criterion, and raw observations are declared and available. The present reported runtime pair supports the stated configuration-specific speedup; it does not by itself estimate runtime dispersion or establish general scalability.

3.3 Computational Dependency Structure

Table **??** makes explicit which assumptions support each computational claim.

4 Benchmark and Reproducibility Registries

The benchmark statuses describe the results reported for the declared configuration. The manuscript reports the recorded runtimes but does not include raw timing samples in the present article package; the planned artifacts below must not be interpreted as already released.

4.1 Benchmark Registry

4.2 Future Benchmark Registry

4.3 Artifact Registry

5 Illustrative Benchmark Confidence Interval (Not a Reported Result)

The present benchmark record contains two measured iterations. It therefore does not support a reliable estimate of runtime dispersion or a confidence interval. The following example is a statistical reporting template only; it is not a result of the present benchmark.

Suppose a future campaign records $n = 30$ independent benchmark runs with sample mean runtime $\bar{T} = 8.250\,\text{ms}$ and sample standard deviation $s_T = 0.420\,\text{ms}$. Then

$$SE = \frac{s_T}{\sqrt{n}} = \frac{0.420}{\sqrt{30}} \approx 0.0767\,\text{ms}. \tag{2}$$

Table 3: Reported benchmark quantities and their evidence basis.

Quantity	**Status**	**Evidence basis**
Workload dimension $N = 1{,}000{,}000$	EMPIRICALLY MEASURED (REPORTED)	Declared benchmark configuration
Nonzero count NNZ = 2,000,000	EMPIRICALLY MEASURED (REPORTED)	Declared CSR workload
Sequential runtime, 8.250 ms	EMPIRICALLY MEASURED (REPORTED)	Recorded time stated in manuscript
Parallel runtime, 3.139 ms	EMPIRICALLY MEASURED (REPORTED)	Recorded time stated in manuscript
Speedup, 2.6282×	DERIVED FROM REPORTED TIMES	Ratio of displayed runtimes
Output agreement	REPORTED AT DISPLAYED PRECISION	Comparison stated in manuscript

Table 4: Benchmark campaigns proposed for future evaluation.

Benchmark campaign	**Status**
Thread-count sweep	PROPOSED FOR FUTURE EVALUATION
Task-threshold sweep	PROPOSED FOR FUTURE EVALUATION
Runtime dispersion and repeated-trial analysis	PROPOSED FOR FUTURE EVALUATION
Multi-platform benchmark	PROPOSED FOR FUTURE EVALUATION
Extended workload scaling	PROPOSED FOR FUTURE EVALUATION

Table 5: Artifacts intended for a future benchmark release.

Artifact	**Status**
`runtime.csv`	PLANNED REPRODUCIBILITY ARTIFACT
`raw_timings.csv`	PLANNED REPRODUCIBILITY ARTIFACT
`memory.csv`	PLANNED REPRODUCIBILITY ARTIFACT
`benchmark_config.yaml`	PLANNED REPRODUCIBILITY ARTIFACT
`environment_report.md`	PLANNED REPRODUCIBILITY ARTIFACT

Using $t_{0.975,29} \approx 2.045$, the margin of error is approximately $0.157\,\mathrm{ms}$, giving the illustrative interval

$$CI_{95\%} \approx [8.093,\ 8.407]\,\mathrm{ms}. \tag{3}$$

A real report must specify warm-up policy, run independence, hardware and software configuration, raw timing observations, the estimator used, and whether runs were paired. The hypothetical mean and standard deviation above must not be mistaken for observed values from this manuscript.